Sankalp Jha

Open Source Developer | Software Engineer
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Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	**7.25/10**
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	**91.2%**
2017 – 2020	High School (X)	SJN International School, Coimbatore	**92.6%**

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (***May 2025 – Sept 2025***)
SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4 *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- **Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- **Validation & Optimization:** Engineered a Jinja2-based P4Runtime testing pipeline (Tofino/Mininet) improving deployment reliability by **30%**, and optimized SID recirculation to achieve **10** Gbps line-rate throughput with **20%** lower inference latency.

Open Source Contributions

- **Project-HAMi/HAMi:**
  #2292 Corrected hardware allocation logic for Kunlun and AWS Neuron AI accelerators to prevent invalid state generation
  #2307 Eliminated node-level deadlocks by strictly enforcing lock releases during failed device-binding operations
  #2259 Stabilized cluster scheduling by adding safe state handling for deleted ResourceQuota tombstones
  #2204 Prevented metrics panics by guarding hardware memory ratio calculations against unknown total memory states
  #2221 fix(workflow): skip non-translatable issue comments
- **WasmEdge/WasmEdge:**
  #4470 fix(wasi): resolve **100%** CPU usage on macOS due to clock mismatch
  #4440 Upgrade Alpine base to **3.23** and restore static LLD support
  #4466 feat(docker): add standalone alpine-base image build pipeline
  #4452 docs(README): fix broken introduction and blog links
- **facebook/bpfilter:**
  #507 build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c

Key Projects

Cutable | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* (*Project Link*)

Objective: Generate editable React applications from text and visual requirements.

Approach: Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.

Impact: Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (*Project Link*)

Objective: Determine whether a startup idea already exists by comparing it against more than **500,000** company records.

Approach: Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.

Impact: Raised USD **10,000** and serves more than **100** users worldwide.

XNIC v1 | *C, Linux Kernel, PCI, DMA, NAPI, QEMU, DPDK (rte_ethdev)* (*Project Link*)

Objective: Build and qualify a defensible clean-room Linux Ethernet driver with observable packet paths, ownership invariants, fault recovery, and a gated user-space DPDK forwarding path.

Approach: Implemented PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX queue backpressure, ethtool counters, and serialized watchdog/reset recovery in C; added staged fault injection, KFENCE and sparse qualification, reproducible QEMU/HVF automation, Wireshark-compatible captures, a W5500 SPI driver path, and an rte_ethdev L2 forwarder.

Impact: Qualified **10,100** RX and TX ring wraps across **646,400** packets at zero loss, **1,000** interface cycles, **100** driver rebinds, and **100** reset-under-traffic cycles; validated DPDK partial-TX cleanup and signal-safe shutdown with the net_pcap virtual PMD.

Do-It | *Go, SSE, HTTP, Raspberry Pi, Local-first* (*Project Link*)

Objective: Provide private, synchronized notes and media across devices on a local network.

Approach: Packaged a lightweight Go SSE and HTTP server for Raspberry Pi, routers, and repurposed phones.

Impact: Delivers self-hosted, account-free synchronization without sending household data to a cloud service.

ebpf-mcp-tracer | *Python, bpftrace, eBPF, Linux, MCP* (*Project Link*)

Objective: Let AI debugging agents inspect Linux performance without granting unrestricted eBPF execution.

Approach: Implemented a strict probe allowlist, dry-run validation, and execution timeouts around bpftrace; exposed the guarded tracing workflow through MCP.

Impact: Enables kernel-level diagnostics while reducing command-injection and runaway-probe risk.

Los-Tecnicos | *Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32* (*Project Link*)

Objective: Build a decentralized energy grid that tokenizes renewable-energy production on the Stellar ecosystem.

Approach: Built a Go backend and **4** Soroban smart contracts; integrated a Raspberry Pi/ESP32 hardware mesh with the platform.

Impact: Won Asia-Pacific Web3 bounties and received recognition from the Government of India.

Open Claw Lab | *Node.js, OpenClaw, Gemini, Oracle Cloud, systemd, Telegram* (*Project Link*)

Objective: Discover high-signal engineering opportunities while preventing unsupported or stale leads from reaching the user.

Approach: Built a multi-agent discovery and independent-review pipeline around typed lead records; added deterministic freshness, eligibility, contact relevance, relocation evidence, and deduplication gates.

Impact: Delivers an evidence-backed Telegram digest while rejecting guessed contacts, unsupported geography, stale roles, and duplicate leads.

Slix | *React, Go, Azure OpenAI, REST API* (*Project Link*)

Objective: Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.

Approach: Built the frontend in React and the service layer in Go; added Azure OpenAI sentiment analysis for curator reviews.

Impact: Gives viewers a concise representation of a movie's tone before watching.

p4Lens | *React, FastAPI, Docker, React Flow, P4* (*Project Link*)

Objective: Make P4 programs easier to learn and debug through an interactive visualization of their processing pipeline.

Approach: Built a parser for P4-**14** and P4-**16** programs; rendered headers, parsers, controls, and deparsers as interactive React Flow topologies.

Impact: Renders pipeline diagrams in under **200** ms and reduces control-plane debugging effort by approximately **40** percent.

Moodish | *JavaScript, Node.js, OpenAI API, MCP* (*Project Link*)

Objective: Rank meal options from a user's craving, budget, and dietary constraints while keeping the recommendation explainable.

Approach: Built intent extraction and deterministic ranking around an AI-generated summary; added a recommendation trace and confirmation-gated cart preview.

Impact: Produces safer, inspectable food recommendations instead of opaque model output.

Myprod | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* (*Project Link*)

Objective: Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

Approach: Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking; added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.

Impact: Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

Technical Skills

- **Language:** Python, C, C++, JavaScript, TypeScript, Go, P4
- **Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- **Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)

- **Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (***May 2024 – Dec 2024***)
  - Led a team of **6+** content writers and **4** designers to deliver documentation and event management support.
  - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and X, achieving a **60%** bump in engagement.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (***Nov 2023 – Nov 2025***)
  - Co-led **10+** tech events & anchored the GDSC WOW flagship event at IIIT Delhi, with the **31+** crew members.
  - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," ***2026*** *International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. **2026**.
  IEEE Xplore ↗ DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

Certifications	• **Open Science 101** – NASA • **Data Science for Engineers** – NPTEL • **Enhancing Soft Skills & Personality** – NPTEL • **IIRS Satellite Remote Sensing** – ISRO
Honors & Awards	• **2x** AKTU State Men Singles TT Gold (**2023**, **2024**) • International Robotics Olympiad Bronze, Thailand (**2019**) • Merit Certification at Carnic Model United Nation (**2021**) • NCC Cadet, National Cadet Corps (**2019**)